

Knowledge Graphs

Lecture 5 - Knowledge Graph Applications

5.2 Knowledge Graphs

Prof. Dr. Harald Sack & Dr. Mehwish Alam

FIZ Karlsruhe - Leibniz Institute for Information Infrastructure

AIFB - Karlsruhe Institute of Technology

Autumn 2020



Knowledge Graphs

Lecture 5: Knowledge Graph Applications

5.1 Ontologies in Action

5.2 Knowledge Graphs

5.3 RDF and OWL Knowledge Graphs

5.4 Knowledge Graph Programming

5.5 Knowledge Graph Visualization

5.6 Knowledge Graph Analytics

How to get Information?

Find a solution for the following question:

- **When was the greenhouse effect discovered?**



Find a solution for the following question:

- **When was the greenhouse effect discovered?**

Google

When was the Greenhouse effect discovered?



All

Images

News

Videos

Maps

More

Settings

Tools

About 14.000.000 results (0,50 seconds)

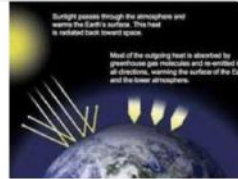
1896

The greenhouse effect was discovered more than 100 years ago

In **1896**, the world renowned Swedish scientist and Nobel Prize Winner Svante Arrhenius (**1859-1927**), described how CO₂ influences the climate. Jun 5, 2018

phys.org › Earth › Environment

80 years since the first calculations showed that the Earth was ...



About Featured Snippets Feedback

People also ask

Who first discovered greenhouse effect?



What is greenhouse effect and global warming?



What is the cause of greenhouse effect?



Feedback

en.wikipedia.org › wiki › Greenhouse_effect

Greenhouse effect - Wikipedia

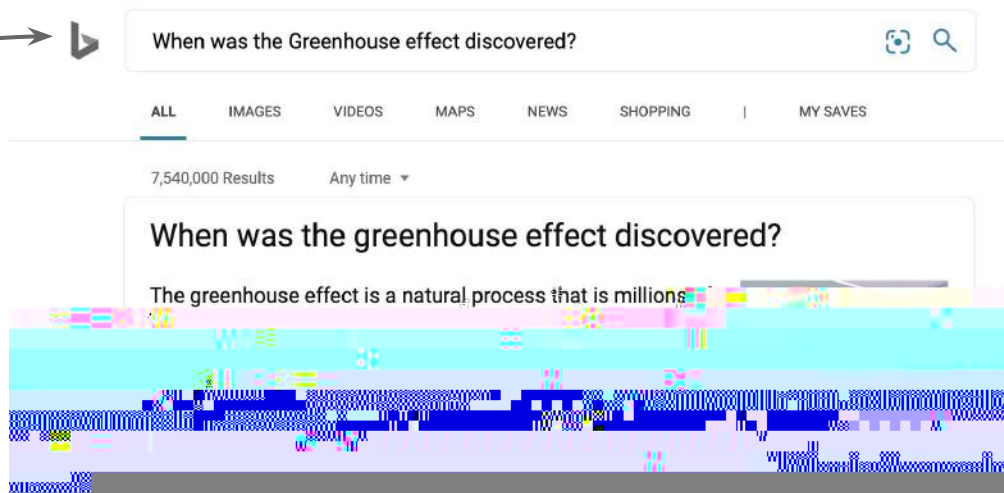
The **greenhouse effect** is the process by which radiation from a planet's atmosphere warms the planet's surface to a temperature above what it would be without ...

[Runaway greenhouse effect](#) · [Anti-greenhouse effect](#) · [History of climate change](#)

www.lenntech.com › greenhouse-effect › global-warming-history

History of the greenhouse effect and global warming - Lenntech

History of the **greenhouse effect** and global warming. ... It was also **discovered** that water vapor absorbed totally different types of radiation than carbon dioxide.



Find a solution for the following question:

- **When was the greenhouse effect discovered?**

How to get Information?

Refine your information need and find out more:

- **What is the greenhouse effect?**

- Refine your information need:

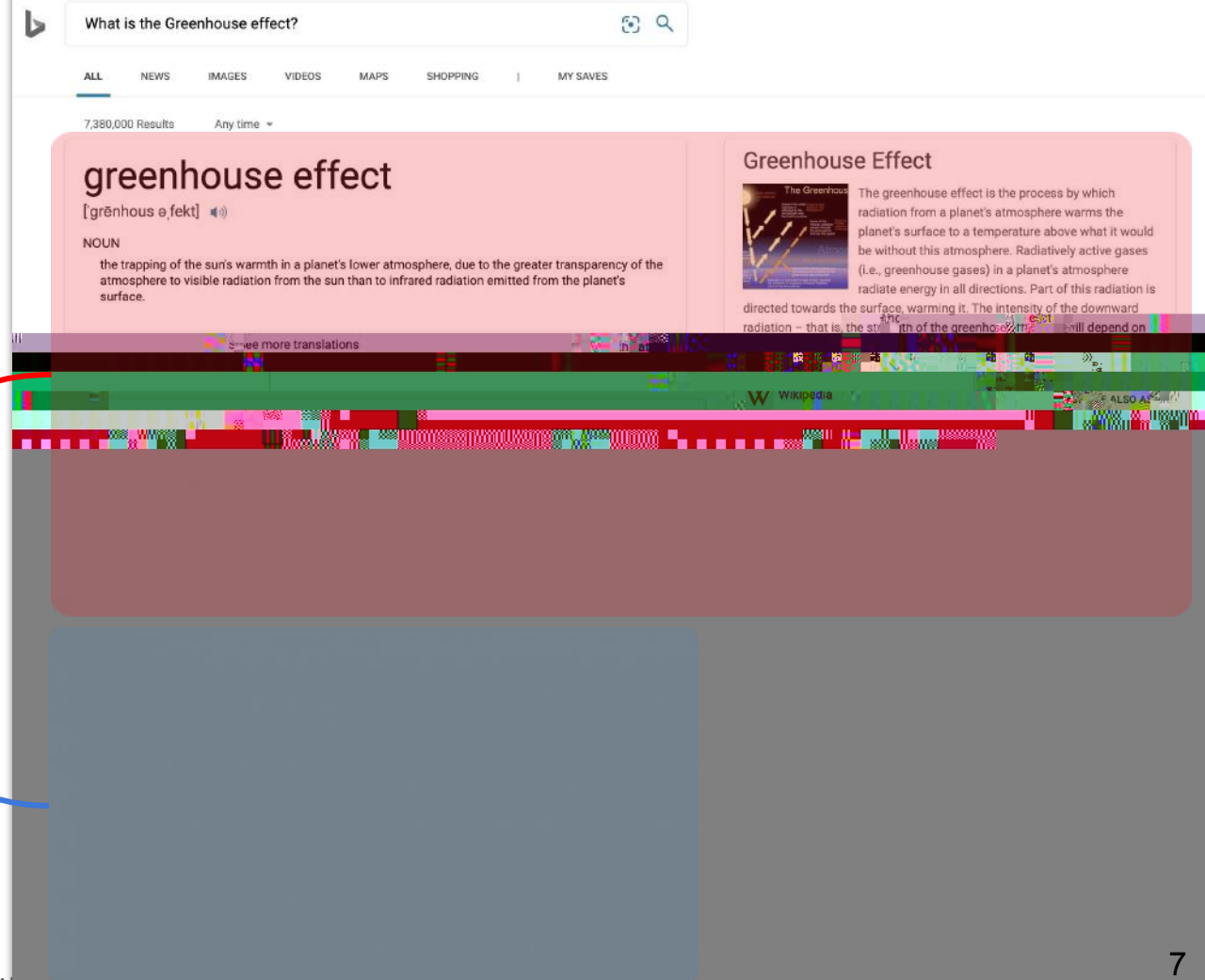
- **What is the greenhouse effect?**

Knowledge Graph

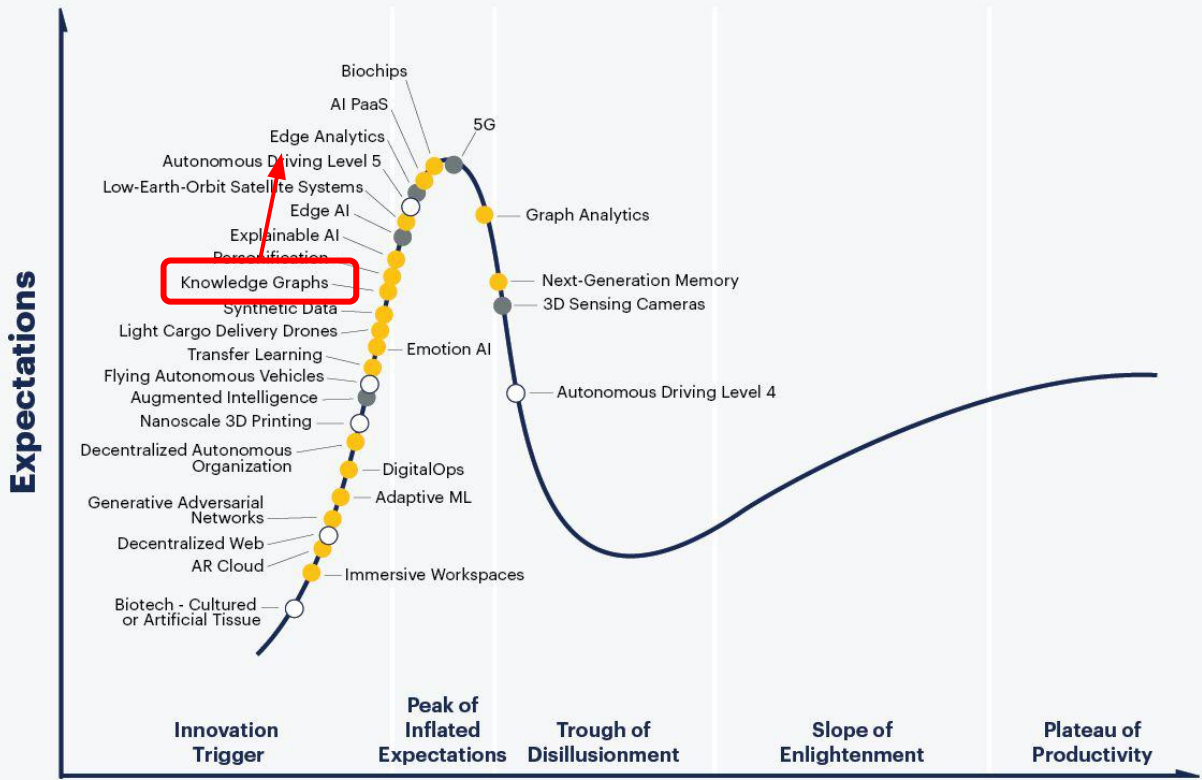
- Facts (structured data)

Search Engine Indexes

- Web pages
- News articles
- Tweets



Gartner Hype Cycle for Emerging Technologies, 2019



Plateau will be reached:

- less than 2 years
- 2 to 5 years
- 5 to 10 years
- more than 10 years
- obsolete before plateau

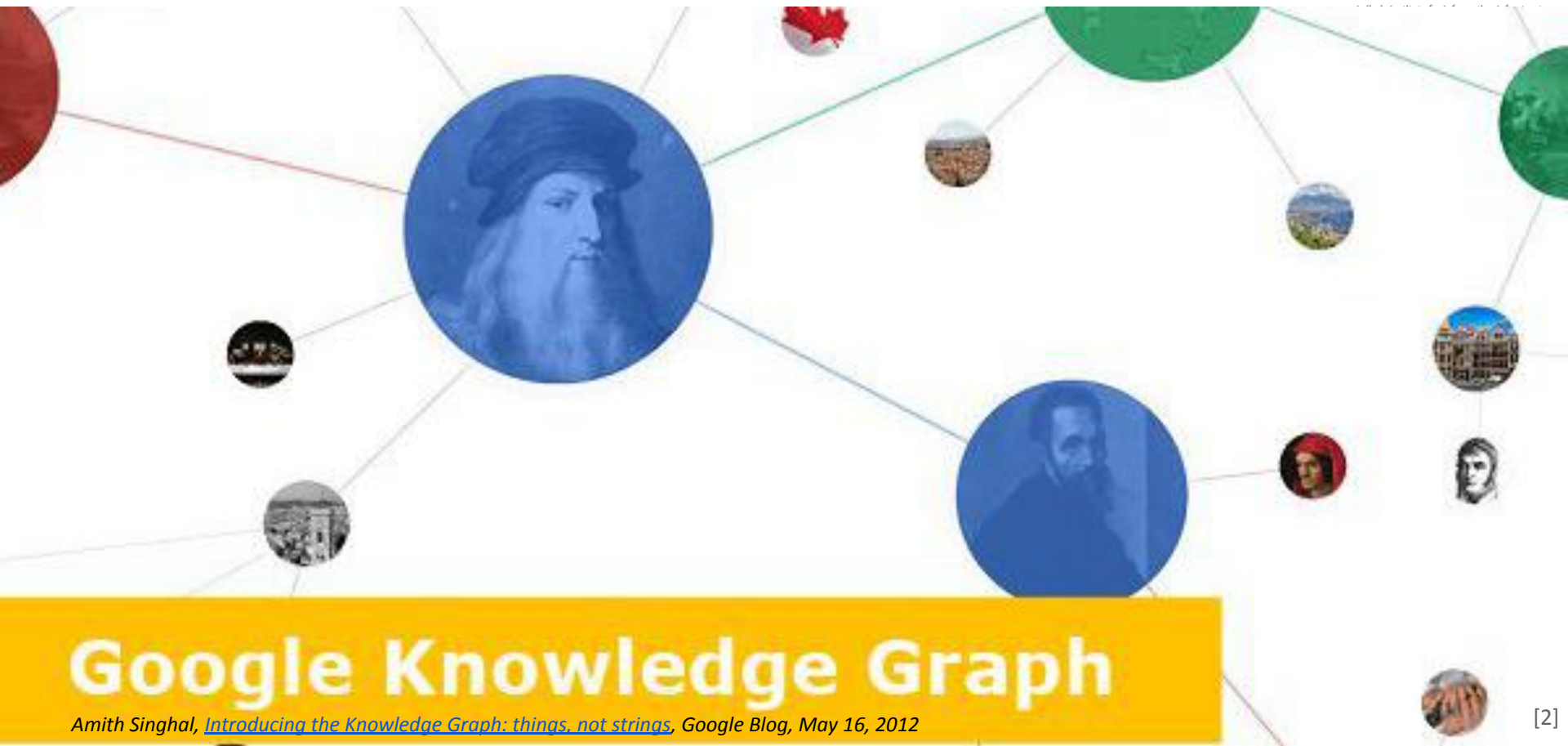
As of August 2019

gartner.com/SmarterWithGartner

Source: Gartner
© 2019 Gartner, Inc. and/or its affiliates. All rights reserved.

[1]

Are Knowledge Graphs Relevant in Practice?



Are Knowledge Graphs Relevant in Practice?

Google

"knowledge graph"



Q All Images News Videos Books More Settings Tools

About 17,700 results (0,14 seconds)



What is a **knowledge graph** and how does one work?

TNW - 11 Jun 2019

Even though it probably affects our lives every single day, most of us have no idea what a "**knowledge graph**" is. Asking your favorite voice ...



Data.world joins forces with Capsenta to bring **knowledge graph** ...

ZDNet - 19 Jun 2019

In a nutshell, data.world uses the superpowers of **Knowledge Graph** (aka Linked Data, aka Semantic Web) technology to integrate datasets and ...

data.world acquires Capsenta to boost **knowledge graph** and data ...

GlobeNewswire (press release) - 19 Jun 2019

[View all](#)



Ontotext's GraphDB 8.10 Makes **Knowledge Graph** Experience Faster ...

PRNewswire (press release) - 13 Jun 2019

Get your GraphDB now and start developing your own **knowledge graph** for advanced search and analytics. About Ontotext. For over two ...

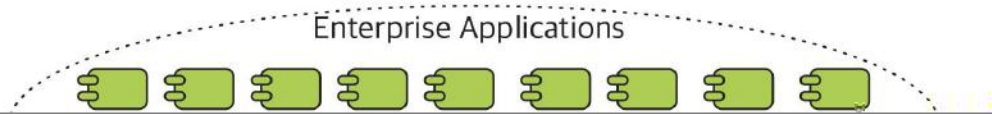
Ontotext's GraphDB 8.10 improves **knowledge graph** experience

KMWWorld Magazine - 12 Jun 2019

Ontotext has released the latest backward compatible update 8.10, of its **Knowledge Graph** semantic graph database. GraphDB 8.10 now ...

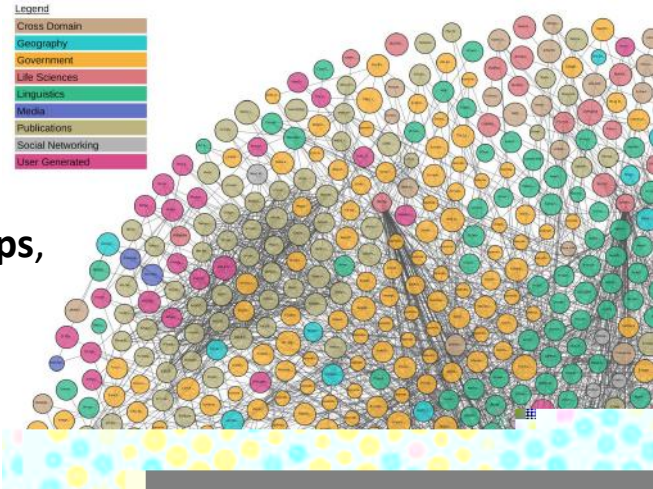


Enterprise Knowledge Graphs



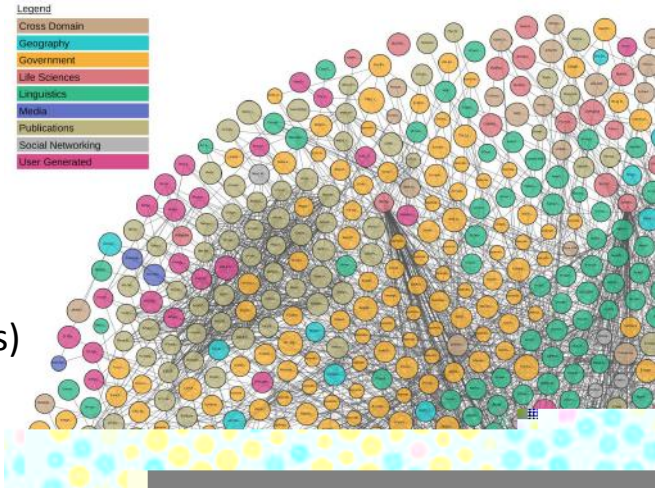
Knowledge Graphs - A Definition

- A **Graph** consisting of **concepts, classes, properties, relationships,** and **entity descriptions**
- Based on **formal knowledge representations** (RDF(S), OWL)
- Data can be **open** (e.g. DBpedia, WikiData), **private** (e.g. supply chain data), or **closed** (e.g. product models)
- Data can be **original, derived, or aggregated**
- We distinguish
 - **instance data** (ground truth),
 - **schema data** (vocabularies, ontologies)
 - **metadata** (e.g. provenance, versioning, licensing)
- **Taxonomies** are used to categorize entities
- **Links** exist between internal and external data
- Including **mappings** to data stored in other systems and databases
- *Fully compliant to **FAIR Data principles** (Findable, Accessible, Interoperable, Reproducible)*

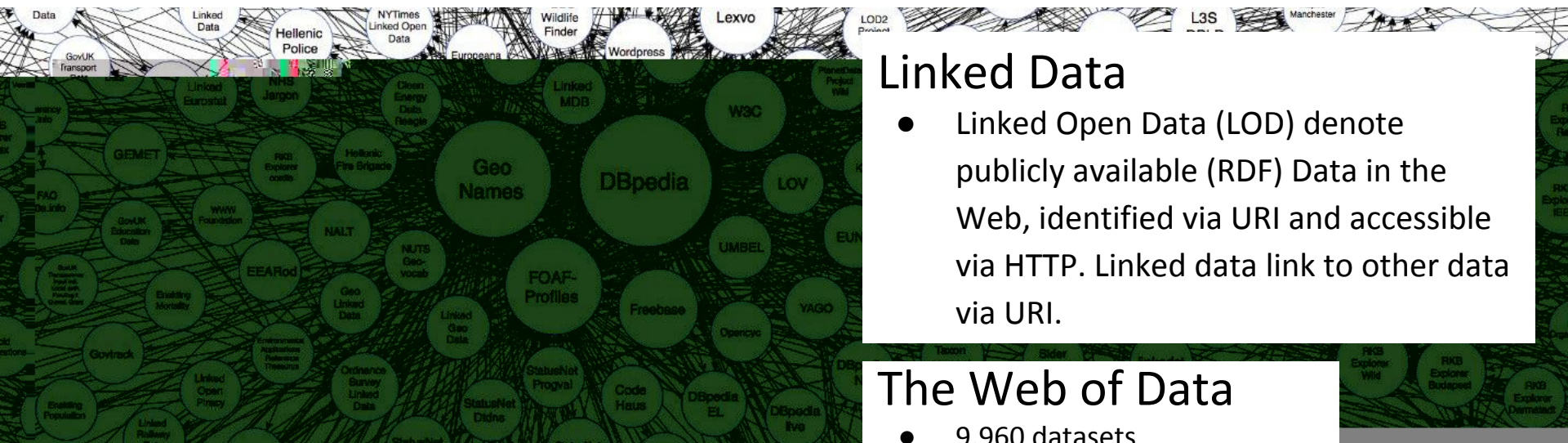


Knowledge Graphs - A Definition (continued)

- A Knowledge Graph is a data set that is:
 - **structured** (in the form of a specific data structure)
 - **normalised** (consisting of small units, such as vertices and edges)
 - **connected** (defined by the - possibly distant - connections between objects)
- Moreover, Knowledge Graphs are typically:
 - **explicit** (created purposefully with an intended meaning)
 - **declarative** (meaningful in itself, independent of a particular implementation or algorithm)
 - **annotated** (enriched with contextual information to record additional details and meta-data)
 - **non-hierarchical** (more than just a tree-structure)
 - **large** (millions rather than hundreds of elements)



The Web of Data - A Web of Knowledge Graphs



Linked Data

- Linked Open Data (LOD) denote publicly available (RDF) Data in the Web, identified via URI and accessible via HTTP. Linked data link to other data via URI.

The Web of Data

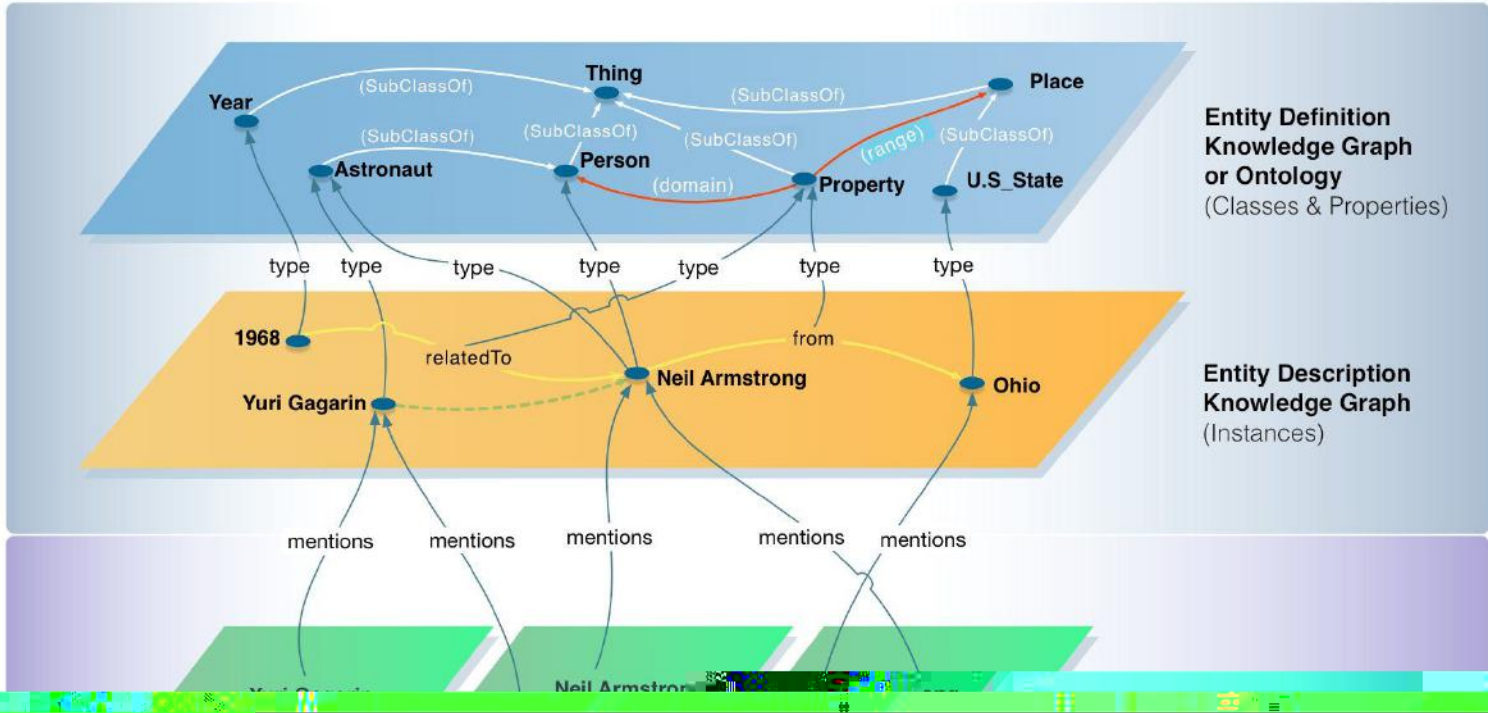
- 9,960 datasets
- >149 billion facts
- >800 million links
(April 2017)

Knowledge Graph — for General Knowledge

HyperLink Artifacts

HyperData

Text



**Entity Definition
Knowledge Graph
or Ontology**
(Classes & Properties)

**Entity Description
Knowledge Graph**
(Instances)

**RDF and OWL
Knowledge Graphs**

Next Lecture...

Picture References:

- [1] Kasey Panetta, The Gartner Hype Cycle highlights the 29 emerging technologies CIOs should experiment with over the next year., 29. August 2019.
<https://www.gartner.com/smarterwithgartner/5-trends-appear-on-the-gartner-hype-cycle-for-emerging-technologies-2019>
- [2] Amit Singhal, Introducing the Knowledge Graph: things, not strings, 16 May 2012, @Google
<https://www.blog.google/products/search/introducing-knowledge-graph-things-not/>
- [3] Fraunhofer IAIS, Enterprise Knowledge Graphs,
https://www.iais.fraunhofer.de/content/dam/iais/fb/SemanticTechnologies/EKG_DIN-lang_Flyer_8seiter_WEB.pdf
- [4] John P. McCrae, The Linked Open Data Cloud, [CC-BY-4.0]
<https://lod-cloud.net/>
- [5] The Linked Open Data Cloud, 2014 version, [CC-BY-4.0],
<https://lod-cloud.net/versions/2014-08-30/lod-cloud.png>
- [6] Kingsley Idehen, Twitter @kidehen, 08.11.19,01:35
<https://twitter.com/kidehen/status/1192601481133875200?s=11>
- [7] Hokusai, The Great Wave off Kanagawa (神奈川沖波裏), from 36 Views of Mount Fuji, (1829-1832), [Public Domain]
https://commons.wikimedia.org/wiki/File:The_Great_Wave_off_Kanagawa.jpg